
Understanding Grammar Representation in the Human Brain via Information-Restricted LLMs

Bachelor's Thesis

Author

Simone Esposito
Matrikelnummer: 469561

Supervisors

Prof. Dr. Fatma Deniz
Prof. Dr. Klaus Obermayer

Berlin, May 2026

Abstract

An integral part of human language is the ability to organize words into grammatical categories such as nouns and verbs, yet how these categories are processed in the brain remains contested. Previous neuropsychological and neuroimaging studies suggested that grammatical class is processed in distinct cortical regions, with nouns associated with temporal areas and verbs with frontal areas. However, these studies usually relied either on brain lesion patterns or used highly controlled stimuli in task-based paradigms, making it difficult to determine whether grammatical class is truly segregated. fMRI data were recorded while participants listened to and read several hours of naturalistic narrative stories. Voxelwise encoding models were constructed using information-restricted contextual embeddings derived from GPT-2. By masking noun and verb tokens during embedding generation, the sensitivity of each voxel to each grammatical class was isolated across participants. Noun- and verb-sensitive voxels were found to be distributed across frontal, parietal, and temporal regions. Verb sensitivity predominated across most language ROIs, while noun sensitivity was concentrated near visual areas. These results suggest that the representation of grammatical class in the human brain is better characterised by a distributed account than by the classic fronto-temporal dichotomy, while also revealing a degree of functional specialisation that challenges a purely uniform distributed view.

Zusammenfassung

Die Fähigkeit, Wörter in grammatikalische Kategorien wie Nomen und Verben einzuteilen, ist ein wesentlicher Bestandteil menschlicher Sprache - wie diese Kategorien jedoch im Gehirn verarbeitet werden, bleibt bislang ungeklärt. Frühere neuropsychologische und neuroimaging-Studien legten nahe, dass grammatikalische Klassen in distinkten kortikalen Regionen verarbeitet werden, wobei Nomen mit temporalen und Verben mit frontalen Arealen assoziiert wurden. Diese Studien stützten sich jedoch meist entweder auf Läsionsmuster oder verwendeten stark kontrollierte Stimuli in aufgabenbasierten Paradigmen, was es erschwerte zu bestimmen, ob grammatikalische Klassen tatsächlich segregiert sind. fMRT-Daten wurden aufgezeichnet, während Teilnehmende mehrere Stunden naturalistische Geschichten hörten und lasen. Voxelweise Kodierungsmodelle wurden mithilfe informationsrestringierter kontextueller Einbettungen aus GPT-2 konstruiert. Durch das Maskieren von Nomen- und Verb-Tokens während der Einbettungsgenerierung wurde die Sensitivität jedes Voxels gegenüber jeder grammatikalischen Klasse über Teilnehmende hinweg isoliert. Nomen- und verb-sensitive Voxel erwiesen sich als über frontale, parietale und temporale Regionen verteilt. Verb-Sensitivität überwog in den meisten sprach-relevanten ROIs, während Nomen-Sensitivität in der Nähe visueller Areale konzentriert war. Diese Ergebnisse legen nahe, dass die Repräsentation grammatikalischer Klassen im menschlichen Gehirn besser durch ein verteiltes Modell charakterisiert wird als durch die klassische fronto-temporale Dichotomie, wobei gleichzeitig ein Grad funktionaler Spezialisierung sichtbar wird, der eine rein gleichmäßig verteilte Sichtweise in Frage stellt.

Introduction

Language is a uniquely human capacity, and grammar is its organizing principle. Among the most fundamental categories of grammar are parts of speech (POS), which encompass various word classes such as nouns, verbs, adjectives and adverbs. Of these, the distinction between nouns and verbs has remained the primary focus of research in psycholinguistics and cognitive neuroscience. To investigate the functional organization of this distinction, researchers have historically relied on neuropsychological case studies of lesion-deficit patterns and, more recently, on brain imaging techniques such as functional magnetic resonance imaging (fMRI).

Neuropsychological studies of patients with focal brain lesions provided the earliest evidence for a **fronto-temporal** dissociation, with temporal regions implicated in noun retrieval and frontal regions in verb retrieval (Damasio and Tranel 1993). Subsequent neuroimaging studies offered partial support for this view, with the left inferior frontal gyrus (LIFG) more strongly engaged by inflected verbs than inflected nouns, suggesting a specific role for frontal regions in processing complex verb morphology (Tyler et al. 2004; Siri et al. 2007). Other accounts have challenged the fronto-temporal dichotomy. The **emergentist view** (Elman 2004) holds that the apparent noun-verb distinction is driven by semantic properties rather than grammatical class per se, with nouns being more imageable and associated with objects and verbs being less imageable and associated with actions (Bird et al. 2000; Vigliocco et al. 2011). The **combi-**

natorial view (Garrett 1984; Ullman et al. 1997) attributes the distinction to morphosyntactic processing demands, pointing to evidence that patients with noun and verb deficits show similar difficulties with pseudowords belonging to the respective grammatical class, suggesting that form and syntactic position rather than meaning drive the distinction (K. Shapiro and Caramazza 2003). A theory-blind meta-analysis of 36 fMRI and PET studies found little consistency across results, with only three regions of interest consistently associated with noun processing and only one consistently linked to verb processing, leading to the conclusion that noun and verb representations are broadly distributed across frontal, parietal, and temporal regions (Crepaldi et al. 2013).

However, these studies relied on highly controlled experimental paradigms, such as picture naming (Damasio and Tranel 1993; Siri et al. 2007), pseudoword production (K. Shapiro and Caramazza 2003), or single-word and simple sentence stimuli. Moreover, lesion-based methods are inherently limited by the diffuse nature of brain damage, making it difficult to isolate the contribution of specific regions to noun and verb processing. These controlled settings do not capture the complexity of grammatical class processing during natural language use (Hamilton and A. Huth 2018). To our knowledge, the representation of grammatical class across listening and reading modalities has not been studied using naturalistic stimuli at the level of individual participants.

In this thesis, I used fMRI data recorded while 9 participants listened to and read the same naturalistic narrative stories from *The Moth Radio Hour* (Deniz et al. 2019). I then used voxelwise encoding models (VEM) to characterize the grammatical class sensitivity of each voxel in each participant and stimulus modality. To isolate noun and verb information, I created information-restricted embeddings using GPT-2 by selectively masking noun or verb tokens during contextual embedding generation. I then compared the predictive performance of the noun-masked and verb-masked models across the cortex to identify regions sensitive to each grammatical class.

Materials and Methods

Stimuli

The speech stimuli consist of 11 short stories taken from **The Moth Radio Hour**, a storytelling program where true autobiographical stories are told in front of a live audience. These were used in previous papers (Huth et al., 2016), including one in which the response data was gathered (Deniz et al., 2019) providing a naturalistic stimulus.

Of these 11 stories, 10 were used as training data, while one 10-minute story was used as validation data. The latter was played twice during the experiment (in two separate runs), and the responses were subsequently averaged across the two runs.

fMRI Data

The fMRI data used in this thesis were originally collected and preprocessed by Deniz et al. (2019). The data set consists of brain responses from 9 subjects who

Table 1: Theoretical accounts of grammatical class representation: predicted patterns versus observed results.

Theoretical account	Predicted pattern	Observed pattern
Fronto-Temporal View <i>Neuropsychological</i> (Damasio and Tranel 1993)	V>N in frontal regions; N>V in temporal regions	V>N in temporal regions (ATL, PTL); no clean frontal/temporal split.
Emergentist View <i>Object-Action, Imageability</i> (Bird et al. 2000; Vigliocco et al. 2011)	V > N in frontal and parietal regions; N > V in inferior temporal regions.	N > V near visual areas; Similar sensitivity in frontal regions.
Combinatorial <i>Morphology/inflection</i> (Tyler et al. 2004; K. A. Shapiro et al. 2005)	N = V broadly, with V > N specifically in IFG for morphological composition.	Slight V > N in IFG; N > V in IFGOrb.
Distributed <i>Shared network</i> (Crepaldi et al. 2013; Caucheteux et al. 2021)	Noun and verb circuits distributed across frontal, parietal, and temporal regions.	Smaller clusters of noun and verb sensitive voxels across the brain.

both listened to and read the stimuli in separate sessions. Each spoken (and written) story was presented during a separate fMRI scan. The sampling rate was 2.0045 seconds, so 1 Repetition Time (TR) = 2.0045 s. Each scan included 10 seconds (5 TR) of silence both before and after the story. These data were collected during two 3-hour scan sessions that were performed on different days. MRI data were collected on a 3T Siemens TIM Trio scanner at the UC Berkeley Brain Imaging Center.

Each functional run was motion-corrected using the FMRIB Linear Image Registration Tool (FLIRT) from FSL 5.0 (Jenkinson and Smith, 2001; Jenkinson et al., 2002). All volumes in the run were then averaged over time to obtain a high-quality template volume. FLIRT was also used to automatically align the template volume for each run with the overall template, which was chosen to be the temporal average of the first functional run for each participant.

For subject 4, no curvature backdrop was available; flatmaps for this subject are hence omitted from the figures.

Features

Several low level features from Deniz et al. (2019) were used as well as lexical embeddings. In addition, embeddings from an LLM were created.

Low level Features

For the listening modality, Deniz et al. (2019) constructed a list of phoneme-time pairs for each of the 39 English phonemes. This list was then down-sampled to the fMRI acquisition rate. The result is a $(39 \times N_{TR})$ model. In addition, a

single-feature model was provided for the number of phonemes present in each volume.

For the reading modality, Deniz et al. (2019) constructed a 26 feature model by counting the number of times each of the 26 letters in the english alphabet was present within each word. The model was also downsampled to the acquisition rate of fMRI. Moreover, a single-feature model was provided for the number of letters present in each volume, as well as a single-feature model for the variance of word length in each volume.

Finally, two additional feature models were provided for both the listening and reading modality. First, a single-feature model that contains the number of words present in each volume. Second, a single-feature model that contains the number of pauses present in each volume.

Lexical embeddings

In addition to the low-level features, I also used the *English1000* lexical embeddings provided by Deniz et al. (2019).

The *English1000* feature space is a 985-dimensional semantic model based on word co-occurrence statistics (Deerwester et al. 1990; Lund and Burgess 1996; Mitchell et al. 2008; A. G. Huth et al. 2016). A co-occurrence matrix M was constructed with 985 rows corresponding to basic English words (drawn from Wikipedia’s List of 1000 Basic Words) and the columns corresponding to the words selected from a large text corpus comprising Moth Radio Hour transcripts, books from Project Gutenberg, Wikipedia articles, and Reddit comments (A. G. Huth et al. 2016).

Each entry M_{ij} in the matrix represents the number of times the i -th basic word co-occurs with the j -th word in the corpus within a 15-word window. The counts were then logarithmically transformed as $\log(1 + M_{ij})$, and the rows and columns were each z-scored to correct for differences in word frequency.

The feature vector for a given word w is then obtained by looking up the corresponding column in the matrix, resulting in a 985-dimensional vector that captures the semantic relationships of w with the basic English words. These features were used when fitting the model to the listening data. For more details on the construction of the English1000 feature space, see Deniz et al. (2019) and A. G. Huth et al. (2016).

Contextual Embeddings

The main limitation of the *English1000* lexical embeddings is the lack of contextual information in the feature space. With lexical embeddings, the same word will be assigned to the same feature vector regardless of the context it appears in. For example, the word *fine* has a different meaning in the sentences “I am fine” and “I paid a fine”.

To account for this, I made use of *contextual embeddings*, which, as the name suggests, aim to include contextual information in the word embeddings. Such embeddings were introduced by the ELMo Long Short-Term Memory (LSTM)

network (Peters et al. 2018) and refined by the BERT transformer model (Devlin et al. 2019).

Contextual embeddings play a fundamental role in all modern Language Models (LM), and for this thesis I used GPT-2’s embeddings, as it is commonly used in related literature (Schrimpf et al. 2021; Tuckute et al. 2024).

Given an input sequence consisting of the i -th word of a story w_i and its k -length context $C_i = (w_s, \dots, w_{i-1})$ with $s = \max(0, i - k)$, I can generate the contextual embeddings through the following steps.

Tokenization

Before a sentence or word can be embedded in vectors, it needs to be broken down into a set of token IDs. This turns the input sequence into a sequence of tokens $(t_1, \dots, t_L) \in \mathbb{N}^L$.

Initial Embeddings

Upon receiving the token sequence, GPT-2 first converts the tokens into embedding vectors through a *lookup table* with a fixed embedding size $d_{\text{model}} = 768$, then adds positional encodings to generate the initial embeddings. These initial embeddings are non-contextual.

Hidden Layers

The initial embeddings then go through a sequence of layers where they are first weighted using the attention mechanism (Vaswani et al. 2023), then passed through a feed-forward network to enrich them with contextual information from surrounding tokens. This yields a new contextualized embedding of the same dimensionality d_{model} .

Previous studies have found that the middle layers, e.g., Layer 8 for GPT-2 small, are best at predicting brain activity in VEMs (Schrimpf et al. 2021; Jain and A. Huth 2018), and Layer 8 is therefore used throughout this work.

Embedding Pooling

The hidden state $H_8 \in \mathbb{R}^{L \times 768}$ contains the contextualized embeddings for every token in the input sequence. These tokens are then averaged to return w ’s contextualized embedding e_w :

$$e_w = \frac{1}{L} \sum_{i=1}^L h_8^{(i)}$$

After repeating this process for each word in each story, I have an embedding matrix $E_j \in \mathbb{R}^{N_j \times 768}$ with $N_j = |S_j|$ being the number of words in the j -th story. To fit the features to the fMRI data, the contextual embeddings are downsampled onto a feature space $F_j \in \mathbb{R}^{K_j \times 768}$ via a Lanczos filter set to the fMRI Nyquist rate (0.25 Hz) with K_j being the number of sample points (TRs) of the j -th story.

The baseline contextualized embeddings are generated using a context length of 20 words for each word in each story.

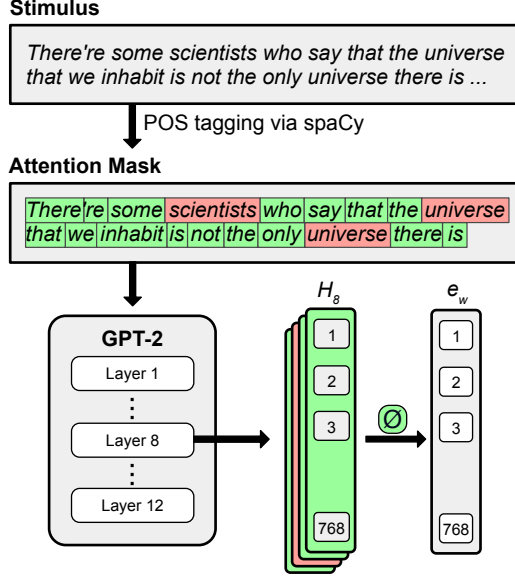


Figure 1. Contextual embedding pipeline with POS attention masking. A narrative stimulus is processed via a part-of-speech (POS) tagger to generate category-specific attention masks. **Green** indicates unmasked tokens, while **red** indicates tokens selectively masked based on grammatical class. The resulting masked input is fed into GPT-2. Hidden states at Layer 8 ($H_8 \in \mathbb{R}^{L \times 768}$) are averaged across unmasked tokens to produce an information-restricted contextualized embedding ($e_w \in \mathbb{R}^{768}$) for each target word.

Information Restricted LLMs

The masking of Parts of Speech (POS) was performed automatically using Python’s `spaCy` library. This library first splits an input sequence into a series of word-like tokens based on language-specific rules. For example, *don’t* is split into two tokens *do* and *n’t*. These tokens are then assigned a POS tag that takes their context into consideration.

Given a set of POS tags P , I can generate a boolean mask for each story. (Figure 1)

$$M_j = (f_{\text{mask}}(t_{j,1}), \dots, f_{\text{mask}}(t_{j,n})) \quad f_{\text{mask}}(t) = \begin{cases} 0 & \text{POS}(t) \in P \\ 1 & \text{POS}(t) \notin P \end{cases}$$

Since proper nouns and auxiliary verbs perform similar functions to nouns and verbs respectively, they are included in the noun and verb POS categories accordingly.

Attention Mask

To apply the POS masks during the generation of contextual embeddings for a word w , the mask values for w and its context C are first mapped to the GPT-2 tokens, yielding M'_w . They are then used as *attention masks*, indicating which tokens the model should attend to and which it should not. The resulting

contextual embedding is calculated by averaging only the unmasked hidden states:

$$\bar{e}_w = \frac{1}{L'} \sum_{i=1}^L m_w'^{(i)} h_8^{(i)} \quad L' = \sum_{i=1}^L m_w'^{(i)}$$

Rather than removing masked tokens from the input sequence — which would alter positional encodings and shift the attention context of remaining tokens — attention masking preserves the original sequence structure while selectively excluding the target POS category from the model’s attention computation.

Voxelwise Encoding Model Framework

Voxelwise encoding models (VEM) aim to predict the BOLD response for each individual voxel v from the stimulus S . This is usually done by first extracting one or more sets of features (called feature spaces) from the stimulus and then modeling the brain response as a linear combination of those feature spaces.

This corresponds to approximating the function $f(S) = g(S)\beta$ where g transforms the N -length stimulus into a $N \times P$ feature space with β as the weights of $P \times V$, V being the number of voxels. This can be accomplished by fitting the feature spaces $g(S_{\text{train}})$ to the responses R_{train} of a training data set, which produces a model $\hat{f}$. The hemodynamic delay is taken into account by including delays of 0, 4, 8 and 16 seconds for each feature, resulting in a finite impulse response (FIR) model.

Finally, the model accuracy is calculated as the per-voxel Pearson’s correlation r_v between the model predictions on a validation dataset $\hat{R}_{\text{test}} = \hat{f}(S)$ and the actual responses R_{test} .

The value of r_v can then be interpreted as the extent to which the response of voxel v to the stimulus can be explained by the given feature space. Projecting r_v onto the cortical surface then yields a sensitivity map to the specific feature space. This shifts the focus from the experimental design to the choice of features, allowing researchers to test different hypotheses using the same data set by varying the feature space.

Banded Ridge Regression

To fit the multiple feature spaces to the BOLD responses, I used banded ridge regression (BRR), which is based on Ridge Regression, a linear regression method that enhances the least-squares objective with a ℓ_2 penalty to reduce overfitting to the training data. This corresponds to finding the weight vector

$$\hat{\beta} = \arg \min_{\beta} \|X\beta - y\|_2^2 + \alpha \|\beta\|_2^2$$

with $\alpha > 0$ being the regularization parameter, which is typically selected by grid search or random search with cross-validation.

BRR extends Ridge Regression by assigning a separate regularization parameter α_i to each feature space X_i (Nunez-Elizalde et al. 2019).

$$\hat{\beta} = \arg \min_{\beta} \left\| \sum_{i=1}^m X_i \beta_i - y \right\|_2^2 + \sum_{i=1}^m \alpha_i \|\beta_i\|_2^2$$

This is important since the feature spaces can vary greatly in dimensionality and predictive power and thus need different levels of regularization.

Results

The goal of this thesis was to determine where the brain represents nouns and verbs and if a theoretical account best characterizes the observed pattern. To answer this question, I generated noun-masked and verb-masked contextual embeddings from 11 stories taken from *The Moth Radio Hour*. These embeddings were obtained by applying attention masks to the GPT-2 input during contextualized embedding generation, selectively suppressing noun or verb tokens. Using banded ridge regression, I fit these features alongside modality-dependent low-level features to the BOLD responses of 9 subjects who listened to and read the stories in separate sessions. Finally, I compared the performance of the noun-masked and verb-masked model across the brain, to identify areas that were more sensitive to nouns and verbs, respectively.

Contextual embeddings predict brain activity across the cortex

To ensure that the contextual embeddings were able to predict brain activity successfully, I first fit the unmasked contextual embeddings to the responses. After fitting the model, I calculated the per-voxel Pearson’s correlation r_v between the model predictions on the validation dataset and the actual responses. The resulting r_v values were then projected onto the cortical surface to produce a sensitivity map to the contextual embeddings (Figure S1).

Contextual embeddings were able to predict brain activity in a large portion of the cortex. The average r_v across statistically significant voxels ($p < 0.05$, FDR corrected) was 0.21 for the listening modality and 0.20 for the reading modality, with some voxels reaching values as high as 0.65. The standard error of the mean (SEM) across subjects was 0.0028 for the listening modality and 0.0021 for the reading modality, indicating that the results were consistent across subjects. This confirms the validity of the contextual embeddings as a feature space for predicting brain activity in response to naturalistic language stimuli.

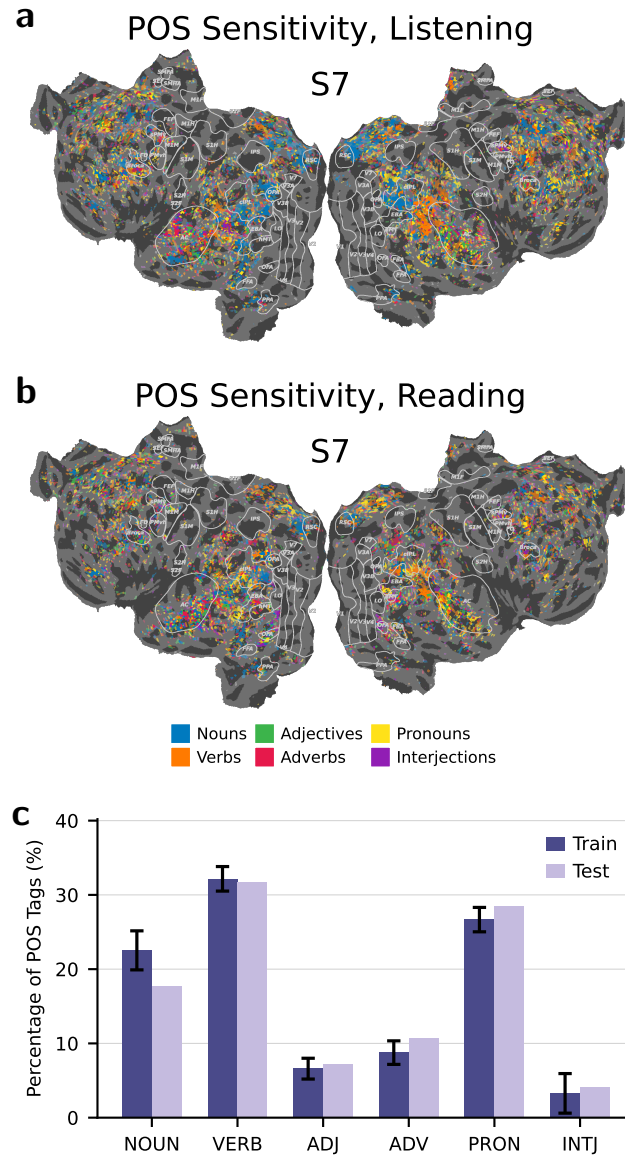


Figure 2. Grammatical category sensitivity across the cortical surface. A voxelwise encoding model (VEM) was used to estimate sensitivity to six parts of speech: nouns, verbs, adjectives, adverbs, pronouns, and interjections. **a**, Drop in predictive performance (r) in the listening modality. Values were computed by subtracting POS-masked model correlations from the unmasked baseline model. Colors represent the category that yielded the greatest accuracy drop for each individual voxel. Voxels whose model prediction was not statistically significant are shown in gray ($p > 0.05$, FDR corrected). **b**, POS sensitivity for the reading modality, computed using the same procedure as in **a**. **c**, Distribution of POS tags in the training and validation datasets. Percentages reflect the frequency of each category relative to the total word count in the narrative stories

POS-masked contextual embeddings reveal clusters of POS sensitivity across the brain

Before focusing on the noun-masked and verb-masked models, I first looked at the performance of a multitude of POS-masked features, including adjectives, adverbs, pronouns, and interjections. For each POS-masked model, the performance drop compared to the baseline unmasked model was calculated by subtracting the r_v values of the masked model from the r_v values of the baseline model. Then, I ranked the drop in performance for each POS category for each voxel individually and plotted the greatest drop on the flatmap in the respective color. This allowed me to identify which areas of the brain were more sensitive to each POS category (Figure 2a,b).

The results showed clear clusters of POS sensitivity across the brain with nouns more prevalent near the Retrosplenial Cortex (RSC) and in parts of the Inferior Parietal Lobule (IPL), while verbs were consistently present near the Temporo-Parietal Junction (TPJ). Nouns and verbs resulted in the greatest drop in performance, followed by pronouns and adverbs, while adjectives and interjections had the least impact on the encoding performance.

The distribution of POS tags in the training stories and the validation story was also analyzed, to verify that the validation story is a representative sample of the training data and that any differences in model accuracy between POS categories reflect the brain’s sensitivity to grammatical class rather than an artifact of unequal stimulus composition. (Figure 2c).

Nouns and verbs show distinct patterns of sensitivity across the brain

To investigate the sensitivity to nouns and verbs in more detail, I focused on the noun-masked and verb-masked models. For each voxel, I calculated the drop in performance compared to the baseline unmasked model for both the noun-masked and verb-masked models. I then plotted the respective drops in performance for each voxel on the flatmap using a bivariate color map, where the blue axis represents the drop in performance for the noun-masked model and the orange axis represents the drop in performance for the verb-masked model. This allowed me to identify areas of the brain that were more sensitive to nouns (points above the diagonal) and areas that were more sensitive to verbs (points below the diagonal), as well as areas that were sensitive to both (grayscale) (Figure 3). The results showed a clear separation between noun-sensitive and verb-sensitive areas.

Noun sensitive voxels appear to be especially concentrated at the margins of the visual cortex, while verb sensitive voxels are more prevalent near the Temporo-Parietal Junction (TPJ). In the frontal region, the clusters appear to be smaller and more interleaved. The distribution of noun-sensitive and verb-sensitive voxels was consistent across subjects and modality, with some variability in the exact location and extent of the clusters.

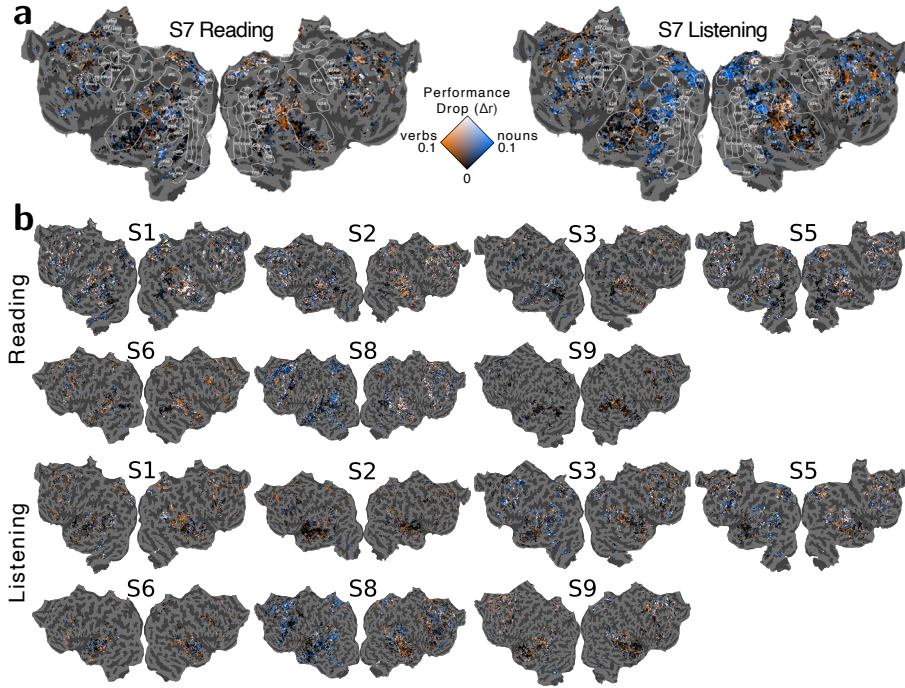


Figure 3. Noun and verb sensitivity across subjects and stimulus modalities. Voxelwise model prediction accuracies (r) for noun-masked and verb-masked contextual embeddings were compared against the baseline contextual model to calculate category-specific drops in performance. **a**, Results for one participant (S7) are shown on the flattened cortical surface for both reading (left) and listening (right) modalities. Sensitivity is plotted using a bivariate color map. **Blue** points indicate a greater drop in performance for the noun-masked model, while **orange** points indicate a greater drop for the verb-masked model. **Grayscale** points indicate a similar drop for both models. Voxels whose model prediction was not statistically significant are shown in gray ($p > 0.05$, FDR corrected). **b**, Noun and verb sensitivity maps for the remaining participants (S1–S9, excluding S4) across both reading and listening modalities. The spatial distribution of noun- and verb-sensitive clusters is largely invariant to stimulus modality.

Noun and verb sensitivity is invariant to stimulus modality

For each subject, I compared the noun and verb sensitivity flatmaps for the listening modality with those for the reading modality (Figure 3). Visually, the noun- and verb-sensitive clusters appear in similar cortical positions across modalities, with noun-sensitive voxels concentrated near visual areas and verb-sensitive voxels near the TPJ.

This result indicates a degree of invariance to stimulus modality, consistent with the finding of Deniz et al. (2019) that semantic representations in this dataset are largely modality-independent. The convergence across listening and reading therefore suggests that the observed grammatical class sensitivity reflects a property of language representation in the brain rather than a modality-specific processing artifact.

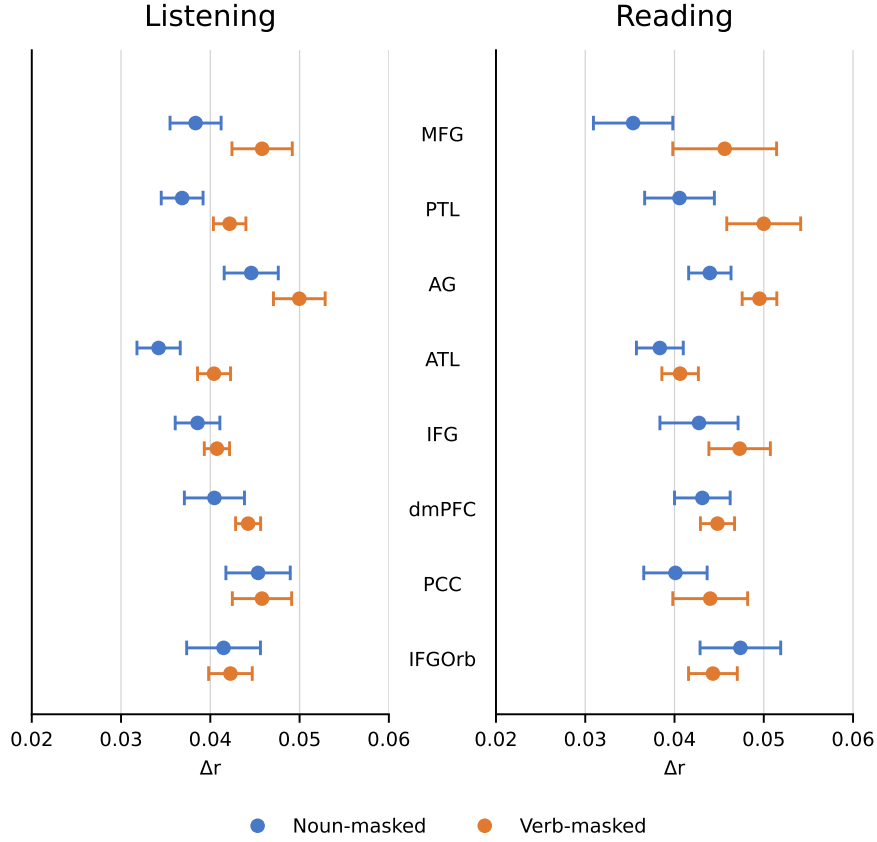


Figure 4. Quantifying noun and verb sensitivity across language-selective regions of interest (ROIs). Each point represents the mean $\Delta r = r_{\text{baseline}} - r_{\text{masked}}$ within significant voxels ($p < 0.05$, FDR corrected) with $\Delta r \geq 0$, for noun-masked (blue) and verb-masked (orange) models, then averaged across participants. Error bars represent ± 1 SEM across eight participants (S1–S9, excluding S4). ROIs are ordered by their mean Δr in the listening modality.

Masking of lexical embeddings produces similar sensitivity maps

To verify whether the apparent clustering was a consequence of the GPT-2 architecture, I constructed noun-masked and verb-masked lexical embeddings by replacing the *English1000* vectors for nouns and verbs with the $\vec{0}$ vector respectively. I then fit the masked lexical features alongside the low-level features to the BOLD responses using banded ridge regression. As in the previous section, I calculated the drop in prediction performance relative to the unmodified *English1000* features and plotted the results using the bivariate colormap (Figure S2).

The resulting flatmaps show similar patterns of noun and verb sensitivity to those observed with GPT-2-based embeddings in the corresponding modality

(Figure 3). Since *English1000* and GPT-2 embeddings represent fundamentally different feature spaces, this comparison is necessarily approximate; nonetheless, the convergence across both models suggests that the observed clusters reflect a genuine neural signal rather than an artefact of the information-restricted LLM.

Verb sensitivity is more prevalent across language ROIs

To further quantify sensitivity to nouns and verbs between different brain regions, I used a set of Regions of Interest (ROIs) based on regions known to be involved in language processing (Negi et al. 2025; Fedorenko et al. 2010). I first mapped each subject to FreeSurfer’s fsaverage template (Fischl et al. 1999). The following ROIs were used: Middle Frontal Gyrus (MFG), Posterior Temporal Lobe (PTL), Angular Gyrus (AG), Anterior Temporal Lobe (ATL), Inferior Frontal Gyrus (IFG), Dorsomedial Prefrontal Cortex (dmPFC), Posterior Cingulate Cortex (PCC), and Inferior Frontal Gyrus, orbital portion (IFGOrb). For each ROI, I calculated the mean performance drop $\Delta r = r_{\text{baseline}} - r_{\text{masked}}$ across all significant voxels ($p < 0.05$, FDR corrected) with $\Delta r \geq 0$, then averaged this quantity across subjects.

The results showed a consistent tendency toward greater verb sensitivity across ROIs, with the mean performance drop ranging from 0.03 to 0.06 for both nouns and verbs for listening and reading. The MFG and PTL showed a consistently larger drop for the verb-masked model across both modalities, suggesting that these regions rely more heavily on verb-specific contextual information. A greater drop in performance for the verbal mask model was also observed in the AG in the reading modality and in the ATL in listening modality (Figure 4).

Discussion

In this thesis, I aimed to investigate the representation of nouns and verbs in the human brain. Using voxelwise encoding models (VEM) and information-restricted LLMs, I was able to create a sensitivity map for nouns and verbs across the brain. The results revealed a pattern of segregated but distributed clusters of noun and verb sensitivity that challenges traditional dichotomies and supports a more distributed account of language processing in the brain.

The observed pattern of noun and verb sensitivity across the brain can be evaluated against the four theoretical accounts: the classic fronto-temporal account, the emergentist account, the combinatorial account, and the distributed account (Table 1).

The fronto-temporal dichotomy is primarily based on neuropsychological studies of aphasic patients and suggests that nouns are processed in temporal areas while verbs are processed in frontal areas. However, the results of this thesis show a higher sensitivity to verbs in temporal areas such as the ATL and AG, which is in direct contradiction to the fronto-temporal account. Moreover, frontal areas such as the IFG, dmPFC and IFGOrbs do not show a clear preference for verbs over nouns, instead showing a mixed pattern of sensitivity.

Therefore, the results of this thesis do not support the classic fronto-temporal account of noun and verb representation.

This discrepancy can be mainly attributed to the difference in methodology. Neuropsychological studies are based on patients with brain damage, which typically extends over large areas and can affect multiple cognitive functions, making it difficult to isolate the specific contribution of a given area to noun and verb processing (Crepaldi et al. 2013). In contrast, VEMs allow for a more fine-grained analysis of the cortical representation of language by fitting specific features to the BOLD response of individual voxels, thus providing a more direct measure of the sensitivity to nouns and verbs across the brain.

The emergentist account posits that the distinction between nouns and verbs is mainly driven by the semantic properties of words, nouns being more associated with objects and verbs with actions (Vigliocco et al. 2011). This view also considers the imageability of the word to play a role in the apparent distinction (Bird et al. 2000). The results show noun sensitive areas in close proximity to visual areas, which would be expected if the imageability of a word were, in fact, an important factor in the distinction of nouns and verbs. However, making definitive conclusions about the role of semantics in the distinction of noun and verb would require investigating the semantic correlates of the observed clusters (similar to A. G. Huth et al. (2016))

The combinatorial account posits that the difference can be attributed to grammatical operations like morphological inflection, which are more common in verbs than in nouns. Indeed, a slightly greater verb sensitivity observed in IFG (Figure 4) could be consistent with this account, as IFG has been implicated in morphological processing (Siri et al. 2007; Yokoyama et al. 2006; Tyler et al. 2004). However, this account does not explain the greater verb sensitivity observed in the ATL and AG, which are not typically associated with morphological processing. Moreover, my methodology does not specifically test for morphological processing, as contextual embeddings capture a wide range of linguistic information beyond morphology, including semantics and syntax.

The distributed account posits that both nouns and verbs are broadly represented in the frontal, parietal, and temporal areas (Crepaldi et al. 2013). The results of this thesis are consistent with this account, as both the noun- and verb-sensitive areas were observed throughout the brain. Particularly in the frontal region, the clusters of noun and verb sensitivity appear to be smaller and more interleaved, which is consistent with a distributed representation. However, the strong separation between noun-sensitive and verb-sensitive areas observed in the parietal and temporal regions challenges a purely distributed account and suggests that there might be some degree of functional specialization for nouns and verbs in these areas. Therefore, the results of this thesis support a more nuanced version of the distributed account, where both nouns and verbs are broadly represented across the brain but with some degree of segregation in certain regions.

The contradictions between the results of this thesis and brain imaging studies can again be attributed to the difference in methodology. The use of naturalistic stimuli and VEMs allows for a more ecologically valid investigation of language processing in the brain, which may reveal patterns that are not appar-

ent in highly controlled experimental settings like block designs or task-based paradigms. In these controlled settings, participants are often required to perform specific tasks that may not reflect natural language processing, and the stimuli are usually limited to single words or simple sentences.

In conclusion, this thesis provides evidence for a complex representation of nouns and verbs in the human brain during the listening and reading of narrative stories, with some degree of segregation in certain regions. This challenges traditional dichotomies and supports a more nuanced view of language processing in the brain that takes into account both distributed representations and functional specialization.

Limitations and Future Directions

In this thesis, I have looked at how the *absence* of nouns and verbs in a sentence affects the voxelwise encoding model prediction performance when compared to the baseline. I used the per-voxel drop in the model’s accuracy as a proxy for that voxel’s sensitivity to nouns and verbs. Future work could investigate whether the *presence* of nouns or verbs in a sentence would produce similar clusters of sensitivity across the brain. This could be achieved using one-hot encodings as a feature space for voxelwise modeling.

The ROIs used for the quantification of the results did not include all verb and noun sensitive clusters seen in the flattened brain maps (see Figure 3), particularly for noun sensitive clusters near visual areas. In addition, many of the qualitatively observed clusters were spread across multiple ROIs. A wider and more granular set of ROIs could be used to better quantify the results.

Although the results appear consistent across both listening and reading modalities, a more rigorous analysis is needed to confirm this consistency.

Beyond addressing these limitations, future research could examine whether these results generalize to other datasets or languages beyond English. Future work might also analyze similarities within the observed clusters beyond grammatical class, such as semantic categories or syntactic properties.

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Supplementary Material

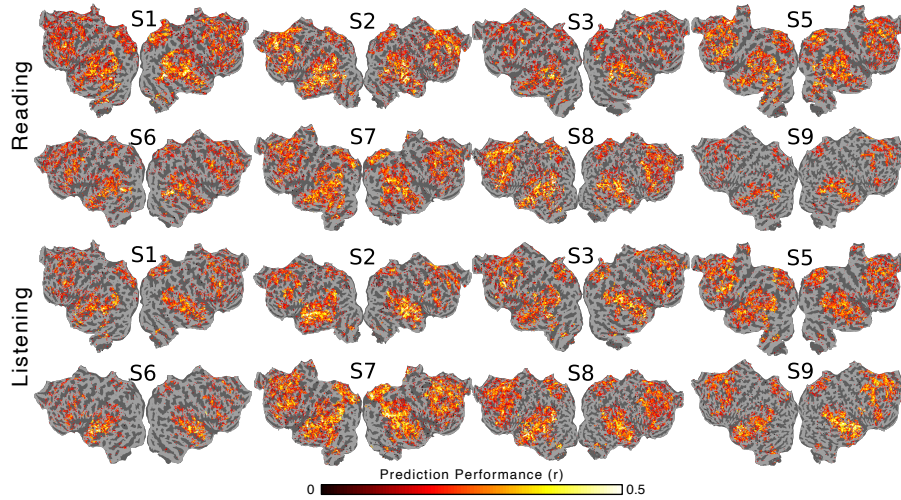


Figure S1. Prediction performance of the baseline contextual model across subjects and modalities. Voxelwise encoding models were used to estimate the relationship between unmasked GPT-2 Layer 8 contextual embeddings and BOLD responses for each individual participant. The top two rows display the model prediction accuracy (r) for the reading modality, and the bottom two rows display the results for the listening modality. Accuracy was computed as the Pearson's correlation between the predicted and actual BOLD responses for a held-out validation story. Each flatmap represents one participant (S1–S9, excluding S4). Voxels whose model prediction was not statistically significant are shown in gray ($p > 0.05$, FDR corrected). This figure demonstrates that contextual embeddings accurately predict brain activity across a large portion of the cortex in both modalities for all participants.

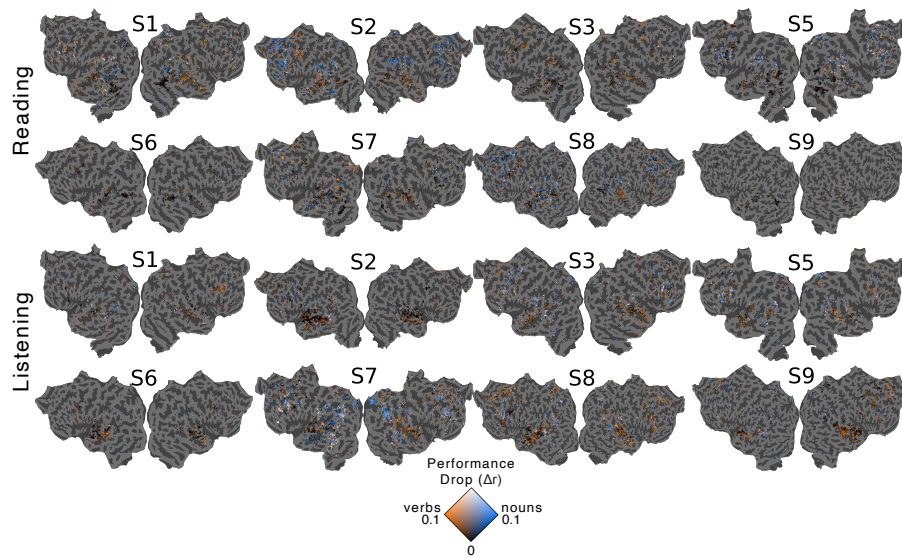


Figure S2. Noun and verb sensitivity across subjects using lexical embeddings. Vox-
elwise encoding models were fit using noun-masked and verb-masked English1000 lexical
embeddings alongside low-level features. The drop in predictive performance relative to
the unmodified English1000 baseline is shown for both reading (top two rows) and listen-
ing (bottom two rows) modalities. Sensitivity is plotted using a bivariate color map where
blue indicates a greater drop for the noun-masked model and **orange** indicates a greater
drop for the verb-masked model. Each pair of flatmaps represents an individual participant
(S1–S9, excluding S4). Voxels whose model prediction was not statistically significant are
shown in gray ($p > 0.05$, FDR corrected). The patterns of sensitivity are similar to those
observed with GPT-2-based contextual embeddings, suggesting that the observed clusters
reflect a genuine neural signal rather than an artifact of the language model architecture.